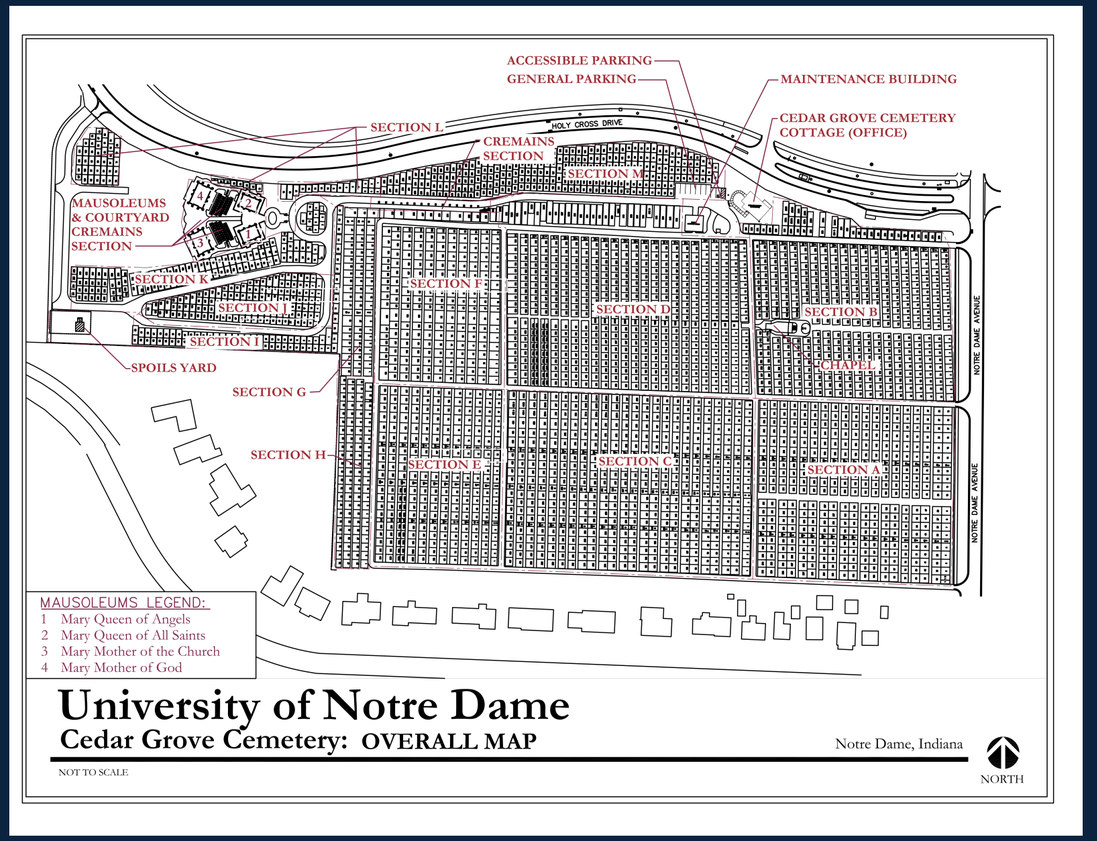
UNIVERSITY OF NOTRE DAME · CEDAR GROVE CEMETERY

The H-Number Project

Turning hand-drawn cemetery plat maps into a searchable digital atlas



Pipeline · Digitized Map · Interactive Site — Status Report

Prepared from the project's decoder, pipeline scripts, and generated data — 2026

EXECUTIVE SUMMARY

Where the project stands today

3,147

lots cataloged across 13 sections

97%

of lots placed on the map (3,063)

185

labels still need a human decision

1

live site — search, records, exports

Every lot in Cedar Grove Cemetery had an old hand-drawn label on paper plat maps — not searchable, not tied to any digital record. This project (1) decoded a new flat numbering scheme (“H-numbers”) for all 3,147 cataloged lots, (2) built an automated pipeline that reads the original hand-drawn labels off the scanned maps and replaces them with H-numbers in place, and (3) published an interactive site where anyone can look up a grave, see what's on record for it, and export section maps or data.

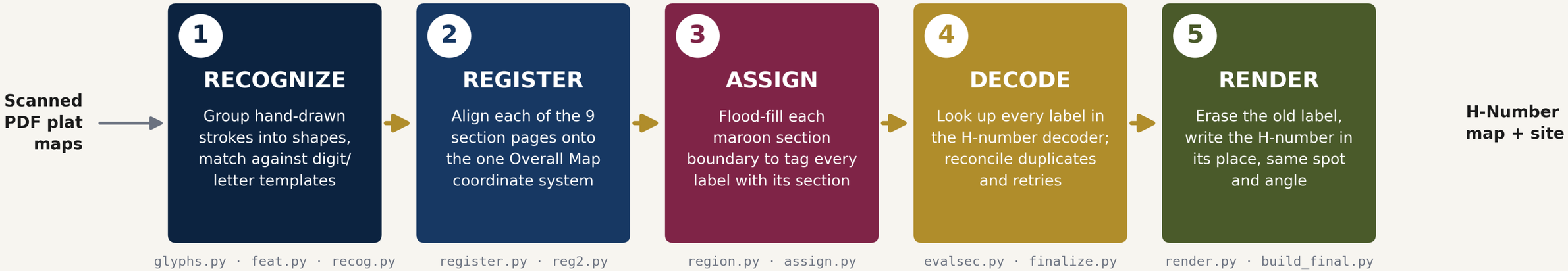
What's left:

136 cataloged lots have no position on the map yet (mostly recognition misses in dense sections); 49 labels read off the drawings match nothing in the decoder's catalog; a handful of numbering disagreements between the decoder and the source drawings need a human call. None of this needs new engineering — it needs someone with local cemetery knowledge to make a handful of judgment calls, detailed on the following pages.

HOW IT'S BUILT

A five-stage automated pipeline

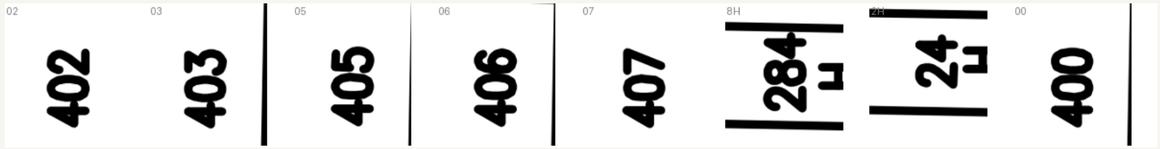
The old lot labels are hand/CAD-drawn line strokes, not text — there is nothing to copy or search. Five stages turn those strokes into placed, searchable H-numbers.



PIPELINE DETAIL · STAGES 1–2

Reading strokes, then placing them correctly

STAGE 1 — RECOGNIZE



Actual glyphs the recognizer read off the drawing — rotated, hand/CAD-drawn digits, no font behind them.

Strokes are grouped into shapes (touching-segment clustering), resampled into a point cloud, and matched against trained digit/letter templates across a rotation sweep — coarse 10° pass, then a refined 2° pass. Each match keeps its top alternatives, so an ambiguous read (9 vs. 6, 1 vs. I) can be resolved later using neighboring context.

STAGE 2 — REGISTER

$$H_{\max}(\text{sec}_i) \longrightarrow H_{\min}(\text{sec}_{i+1}) - 1$$

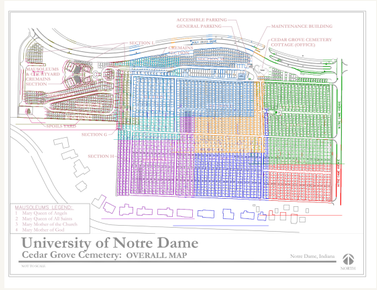
Every section's table starts where the previous one's ends — one gap-free sweep, A→M

The 9 individual section pages and the one Overall Map are separate coordinate systems. A Hough-style vote over matching line segments (angle + length) finds the best-fit scale/translate transform per page, then 8 rounds of nearest-neighbor refinement sharpen it — final residual well under 0.05 map units, so a label's registered position lands essentially exactly on its true spot.

PIPELINE DETAIL · STAGES 3–4

Knowing which section, then decoding it

STAGE 3 — ASSIGN



Every section, flood-filled from its own maroon boundary line — one color per section, from geometry alone.

The maroon section-boundary lines on the source drawing are rasterized and flood-filled (scipy binary_fill_holes), giving every point on the page a section ID — independent of what any label says. This is how a recognized label gets tagged “this one is in Section D” before it's ever looked up.

STAGE 4 — DECODE

$$H(B,\ \text{lot}) = \text{SWEEP} \times (r - 1) + c$$

Section B — formula sweep, column c, row r

$$H_{\text{grave}} = H_{\text{lot}} \times 10 + g \qquad (1 \le g \le 9)$$

Every grave position, all 13 sections

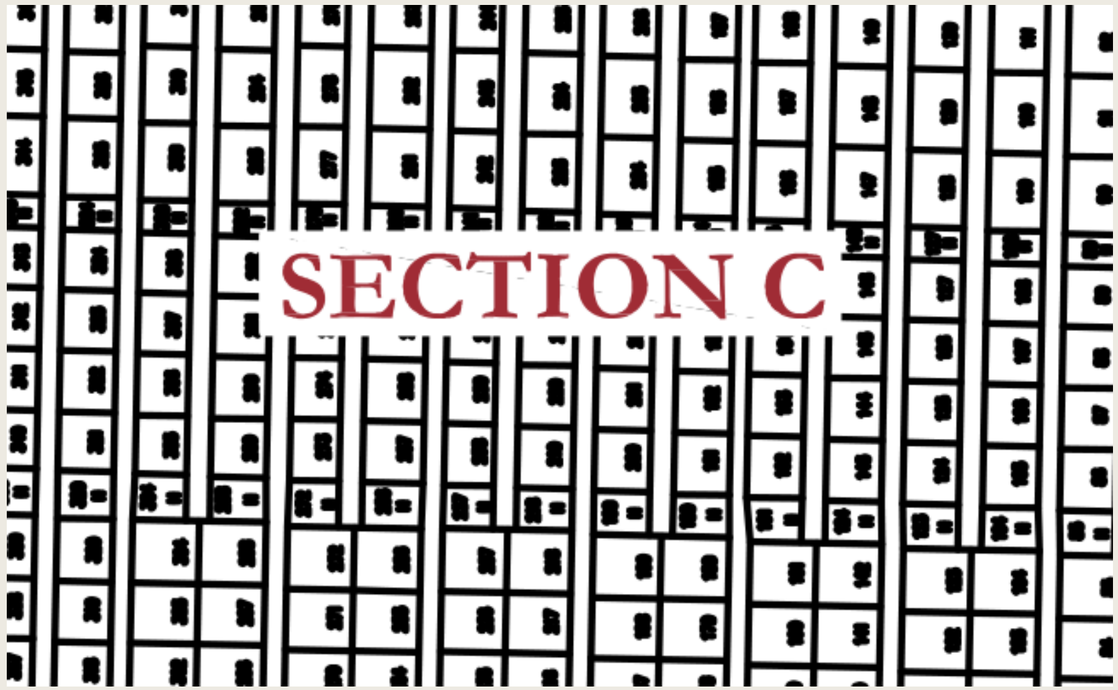
Every recognized label is looked up against s4decoder1.py for its assigned section. Failed lookups get three automatic retries: swap letter/digit order, re-generate the string from each glyph's alternate candidates, or drop one stray glyph and try again — only what survives none of this becomes a genuine open question.

THE RESULT

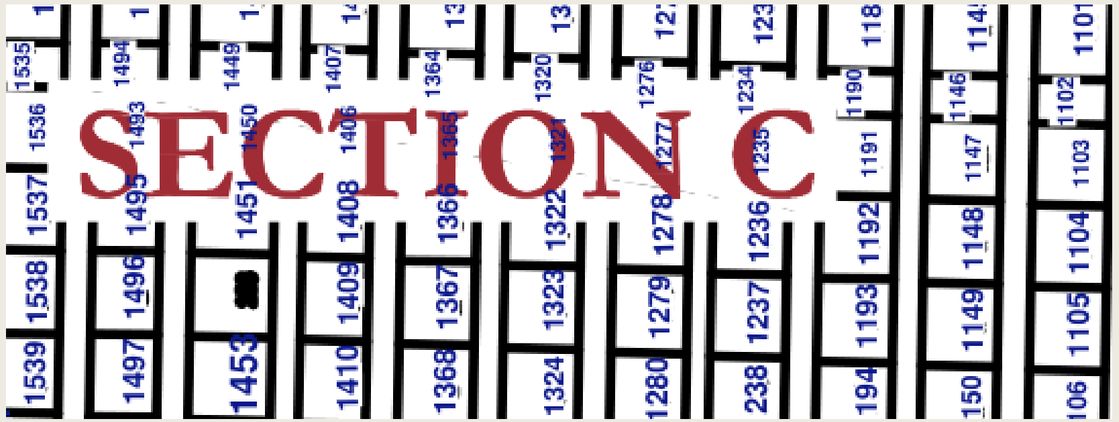
Same spot on the map, before and after

Exact same region of the Overall Map (Section C). The old label is erased and its H-number is written in the same spot, at the same rotation — short form, no leading “H” or zeros (1871 = H001871), matching the size a lot can actually print legibly at.

BEFORE — original hand-drawn labels

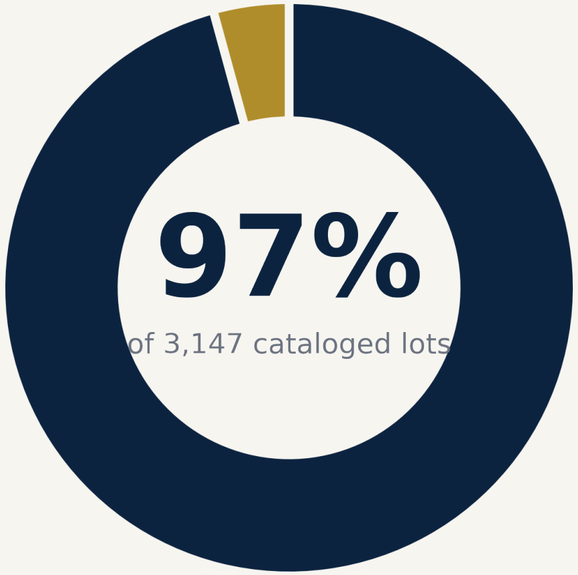


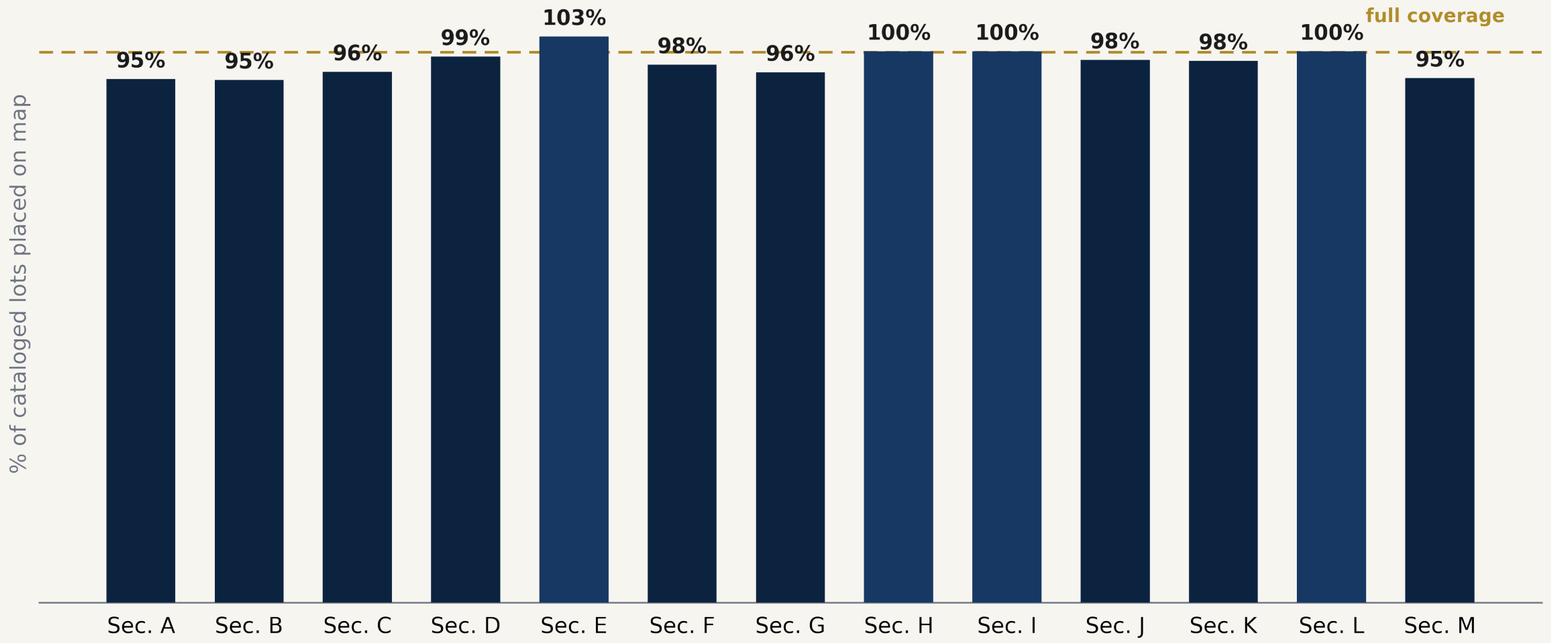
AFTER — H-numbers, same position and angle



RESULTS

97% of the cemetery is placed and searchable

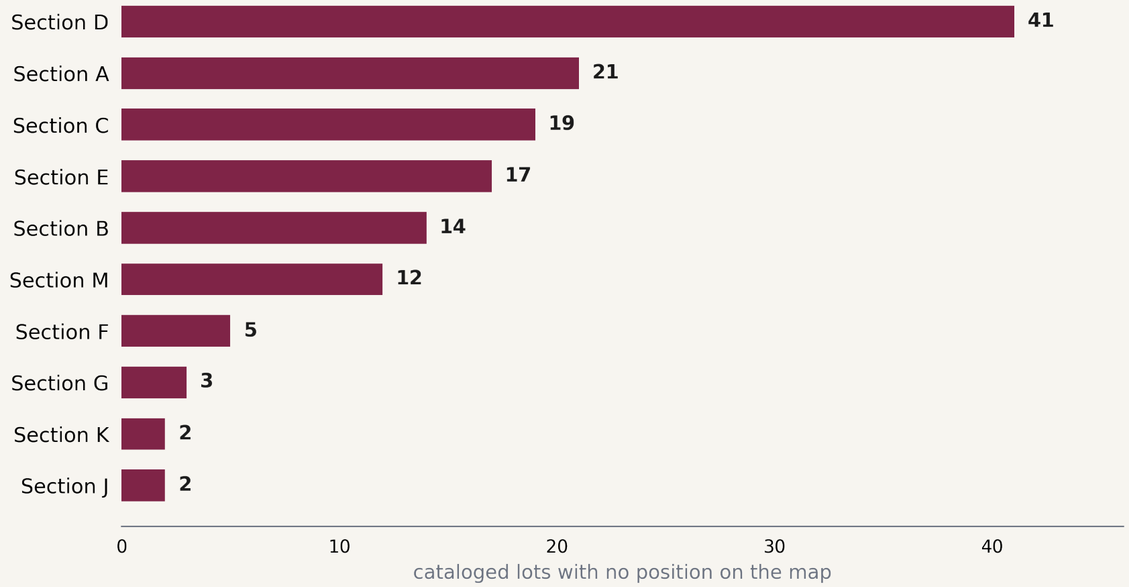




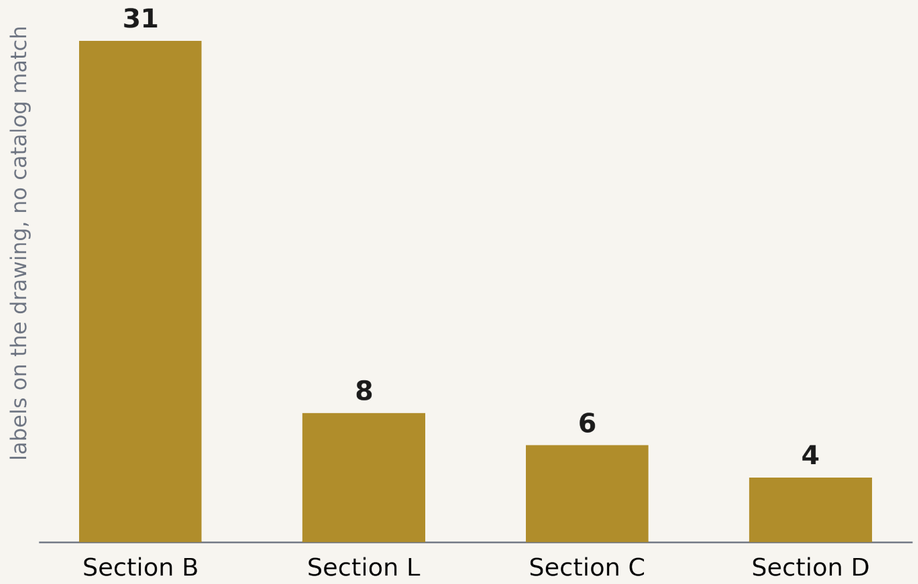
3,063 of 3,147 cataloged lots are placed on the map today. Ten of thirteen sections sit at 95% or higher; four (H, I, J–adjacent, L) are essentially complete.

OPEN ITEMS

Two different problems — kept separate on purpose



Cataloged lots with no position on the map yet (136 total) — mostly recognition misses in dense sections.



Labels read off the drawing that match nothing in the decoder (49 total) — a real open question, not a reading error.

OPEN ITEMS

Three specific numbering disagreements

Section B — label collision

The labels 373–380H exist in two independent tables: the main grid's own formula, and the separate margin-strip table. The margin table is checked first and always wins.

```
h_lot('B','380H') → H002341  (margin)
decode_h(793) → B · 380H  (grid — unreachable)
```

Needs a decision: is the margin authoritative, or do both areas need distinguishing labels?

Section G — duplicate ‘19’

Row 1 of Section G's table lists lot ‘19” twice. The builder keeps the first match; the second still consumes an H-number that can never be reached by a forward lookup.

```
_G_ROWS[1] = ['58','19','19','18', …]
→ 80 reachable lots from 81 row positions
```

Needs a look at the physical drawing to confirm it's a transcription duplicate, then a one-line table fix.

Section A — missing ‘H’ suffix

18 lots (20H, 39H, … 343H) are in the decoder with an H suffix, but the drawing shows a plain number where each should be — the recognizer reads these reliably everywhere else.

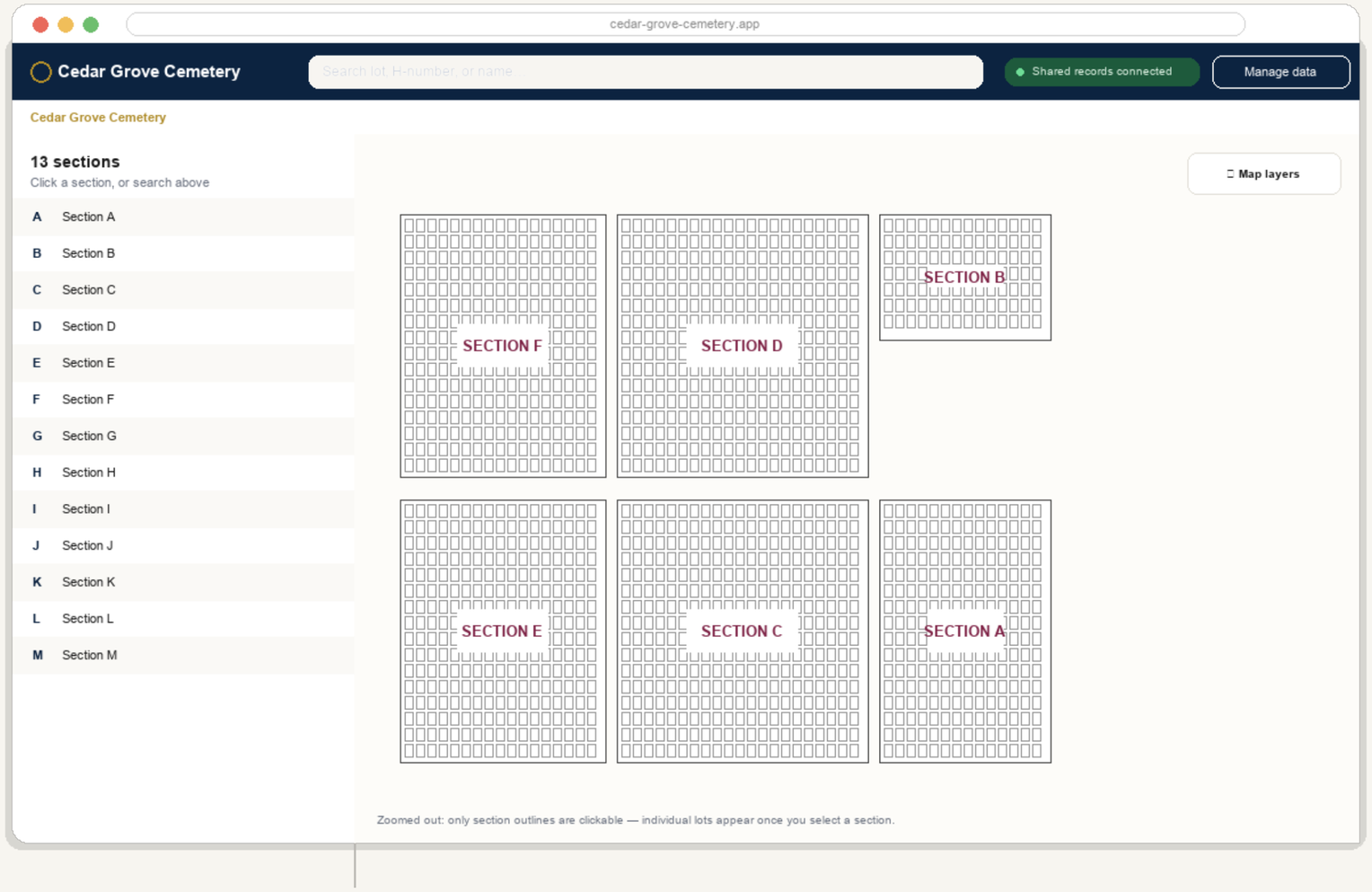
```
Decoder expects: 20H, 39H, 58H …
Drawing shows:   20,  39,  58 …
```

Needs a decision: does the drawing or the decoder's table carry the real intent here?

THE SITE

Cemetery → section → lot → grave

A single interactive page: search or click through the map, and see what's on record. Below are the layout diagrams for each screen (redrawn from the live app's actual design and copy, not screen captures) with the reasoning behind each choice.



Zoomed out = sections only

Individual lot numbers stay hidden until you pick a section — 3,000+ labels at once is noise, not information.

Native vector pan/zoom

The map redraws its own SVG viewBox, not a scaled bitmap — sharp at any zoom, no blur-then-clear delay.

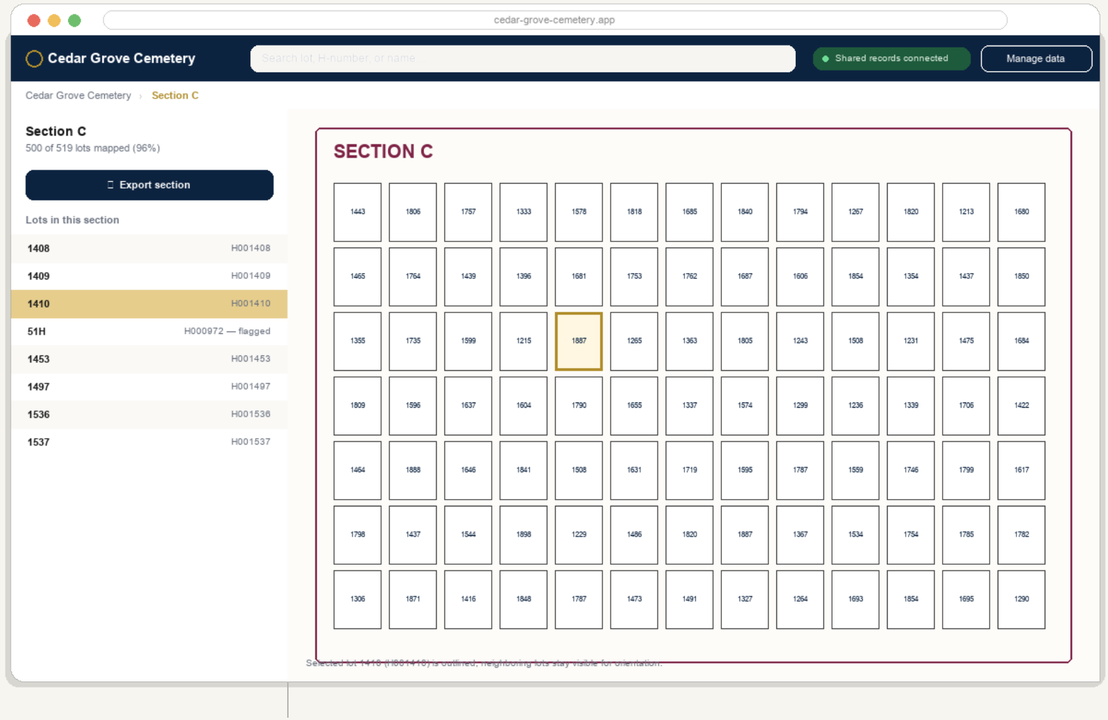
One connection-status chip

States what's live (shared records / read-only / unavailable) up front, not discovered by clicking Save and failing.

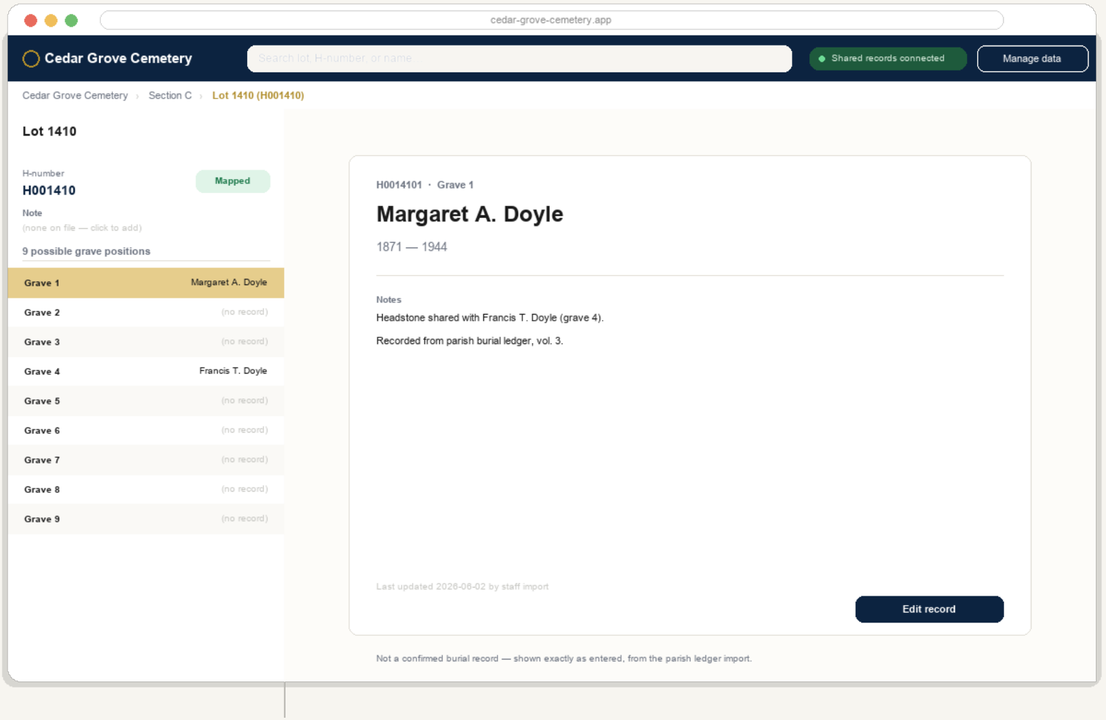
THE SITE

A lot's record, read before it's edited

Section view — every lot listed with its H-number; selecting one outlines its real footprint on the map.

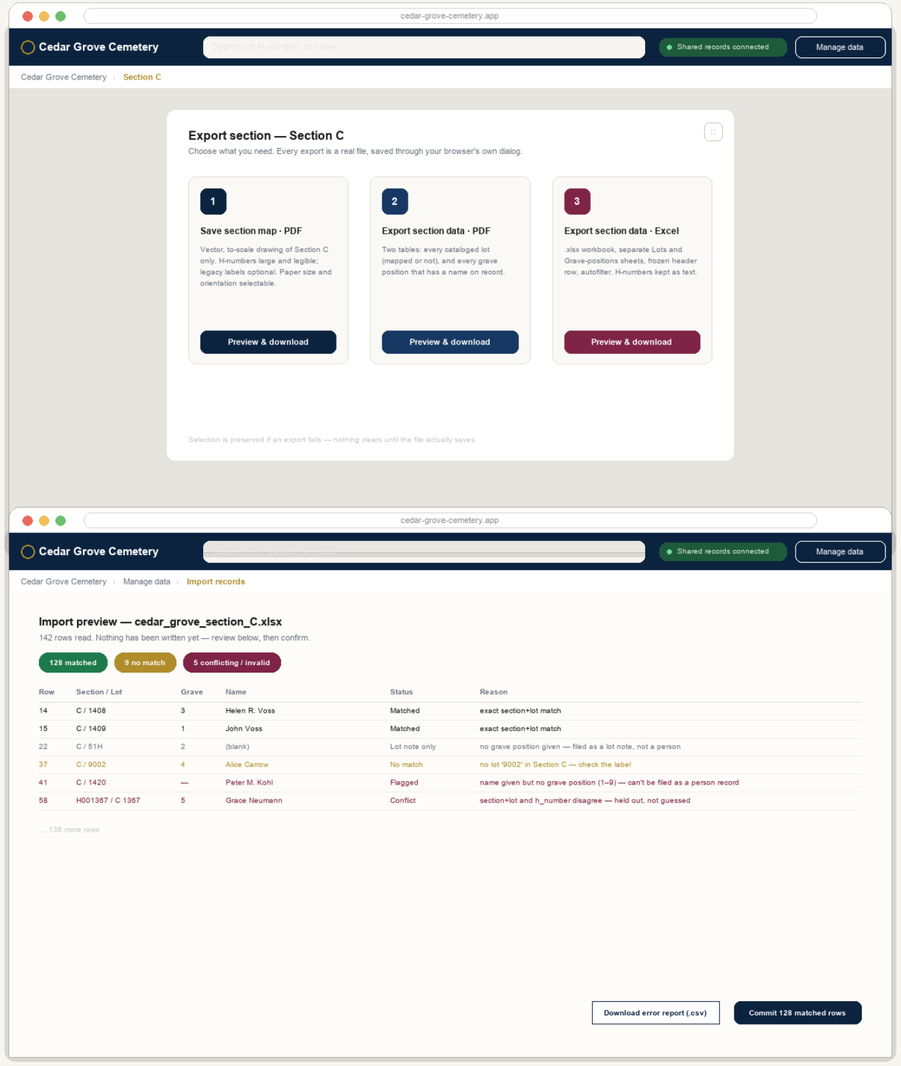


Reading a grave record is a separate step from editing it — no accidental overwrites, and it's explicit that this is not a verified burial.



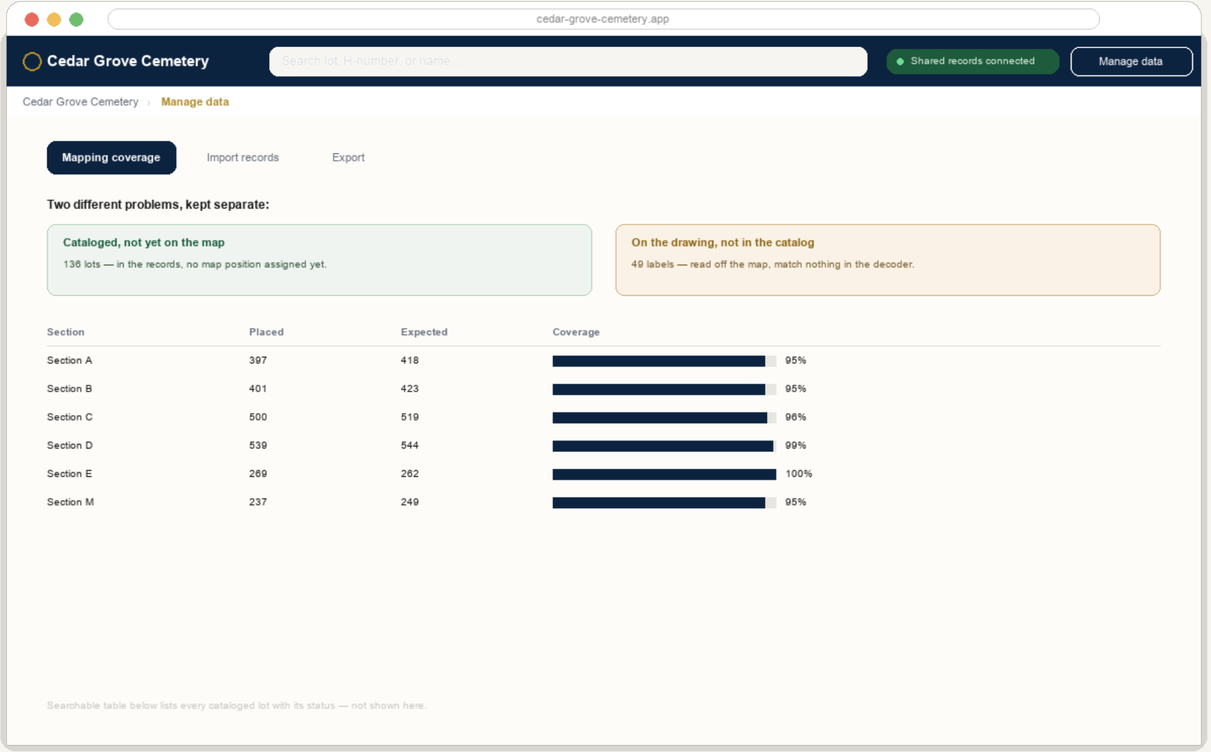
THE SITE

Getting data in, and getting it back out



THE SITE

Two problems, never merged into one list



Why keep them separate?

“Cataloged, no position yet” is a mapping gap on our side. “On the drawing, no catalog match” may be a genuine numbering block someone else needs to explain. Merging them would hide which is which.

Never guesses

No fuzzy matching anywhere in the app — import, search, and this table all use exact matches only.

WHAT'S STILL MISSING

Grave capacity is assumed, not measured

H_grave = H_lot × 10 + g gives every lot 9 possible positions — the formula's ceiling, not a verified fact about any specific lot. Real lots vary: some hold 5 graves, some hold 7. That number has never been recorded anywhere.

Today — grave slots are formula-derived

Every lot gets the same 9 possible positions; H_grave = H_lot × 10 + g

Lot H001410

Grave 1	on record	
Grave 2	no record	
Grave 3	no record	
Grave 4	on record	
Grave 5	no record	
Grave 6	no record	
Grave 7	no record	
Grave 8	no record	
Grave 9	no record	assumed max — not verified

The gap: some lots physically hold 5 graves, some hold 7 —
the real per-lot capacity has never been recorded anywhere.
The formula's "9" is a ceiling, not a fact about any specific lot.

Needed — one master capacity sheet

A single spreadsheet, one row per lot, filled in once and imported

Section	Lot	H-number	Actual graves	Source
C	1408	H001408	6	headstone count, 2024 survey
C	1409	H001409	?	— not yet surveyed
C	1410	H001410	5	plot deed, parish office
D	376A	H001634A	7	field visit, family plot
D	376B	H001634B	?	— not yet surveyed
M	18	H002592	?	— not yet surveyed

How it plugs in — no new feature required:

1. Build this sheet once: section, lot, H-number, true grave count, and where the count came from.
2. Import it exactly like any other spreadsheet — the site's Import records flow already matches by section+lot or H-number.
3. Add one field to each lot's record: "grave capacity." The 9-slot display becomes N-slot, per lot, immediately.
4. Unsurveyed lots keep showing 9 (today's safe default) until their row is filled in — nothing regresses.

EXPORT CAPABILITIES

Every export is a real, saved file

No browser print-to-PDF anywhere — every export goes through the platform's own save dialog, vector PDFs built from scratch, and a real .xlsx workbook (not a renamed CSV).

PDF

Section map

Vector, to-scale, one section. Paper size + orientation choice. Previewed before download.

PDF

Section data

Two tables: every lot (mapped or not) and every named grave position.

XLSX

Section data

Separate Lots / Grave-positions sheets. Frozen header, autofilter, H-numbers as text.

PDF

Location sheet

One lot, its identifiers, and nearby lots for orientation — printed in the field.

PDF / record list

Print list

Any hand-picked mix of lots and grave positions, cemetery-wide, with recorded details.

NEXT STEPS

What's left, and who it needs

1

Resolve the 3 named numbering disagreements

Section B collision, Section G duplicate, Section A missing 'H' — needs someone with cemetery-office context, not more engineering.

2

Review the 49 unmatched drawing labels

Concentrated in Section B (31 of 49) — could be a genuine unlogged numbering block (annex, expansion, renumbering).

3

Spot-check the ~106 real recognition misses

Confirm against the source drawing whether each is a read failure or another decoder/drawing disagreement.

4

Build the grave-capacity master sheet

One row per lot: true grave count and its source. Imports through the existing spreadsheet flow — no new feature.

5

Copy mkdata.py / mksvg.py into pipeline/

Currently only in a scratch workspace — needed there to ever regenerate the site after a decoder fix.